

implemented. This allows individuals to become small-scale manufacturers and sellers.

This not only helps humans but even animals like lizards and penguins since a proper scan can be made with a 3D scanner for 3D modelling. There are many cases already where body parts like prosthetic paws, beaks, and leg braces have helped the suffering animals.

For example, a dog in the US, which was born with a malformation in the legs, can now run very efficiently through 3D-printed wheel-shaped legs. In another instance, a 4-month-old kitten Sonic was missing a bone in one of his front legs, which could have caused extreme deformities in the rest of its body later on. A prototype was developed for the paw, and the movement of the kitten was noted to make

adjustments. Finally, a comfortable paw was made out of carbon fibre, which allowed the cat to move easily.

Be it for humans or animals, it is extremely important to choose the right brand to ensure quality devices. 3D printing provides a promising possibility, but there is a need for specific design guidelines so that the right choices can be made.

—Ayushee Sharma

3 IoT TO PREVENT FOOD WASTE

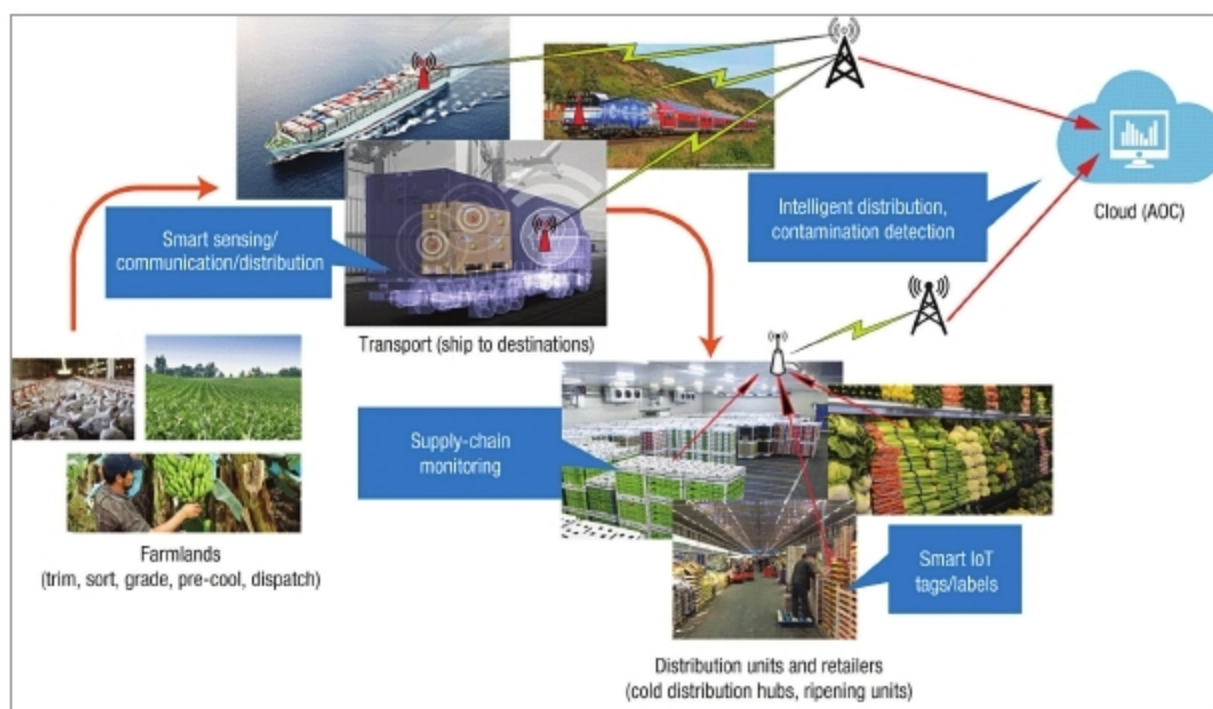
Need for IoT is just not restricted to farms; it also plays a vital role in the entire supply chain. IoT systems have sensors that capture data during transit to keep track of parameters that affect food quality in real time

The Internet of Things (IoT) is playing a great role in the agriculture industry through continuous monitoring of important resources at farms to ensure a good harvest. However, the need for IoT is just not restricted to farms; it also plays a vital role in the entire supply chain, from producers to end consumers.

Companies like Walmart and Amazon procure directly from the farmers to guarantee freshness. But maintaining the freshness of edible items is a challenge. Freshness is measured by shelf life, which starts decreasing from the moment such perishable products are harvested.

With time, as the harvest moves along the supply chain, their quality degrades to such an extent that these products become unsuitable for consumption. The amount of time varies for every item. Quality may decrease at a rapid rate due to environmental factors such as high temperature and humidity. This leads to wastage of food items and loss for retailers. In case such items are still used for consumption, these can potentially cause food-borne diseases.

For this, continuous monitoring is essential. Human-based monitoring is error-prone and tedious. This calls for change at a large scale, which is possible only through IoT. IoT systems have sensors that can act as precise



IoT deployment for entire food chain (Credit: <https://innovate.ieee.org>)

indicators for both businesses and consumers by capturing data during transit to keep track of parameters that affect food quality in real time.

Crest Foods, a restaurant franchisor company, chose refrigeration solutions offered by R9 Technology, a provider of IoT solutions and services for US-based Nestlé Toll House Café by Chip. The main component of the cold-chain monitoring solution is the cellular IoT Gateway, 'Picogate.'

Picogate is designed around Telit Communications' LE866A1-NA cellular (4G) module, LE866 series variant for the North American market. The module enables significant reduc-

tions in cost, power consumption, and size, which makes it useful for the implementation of LTE technology in IoT, especially in machine-to-machine (M2M) communication.

deviceWISE IoT platform by the company comes with a broad library of pre-built device drivers, connectivity management tools, and application connectors to manage several devices and ensure efficient cellular connectivity. Wired and wireless sensor technologies send data over a sub 1GHz local wireless network to Picogate. deviceWISE IoT platform easily connects to a cloud refrigeration monitoring solution, which has a simple user